

# Artificial Intelligence & Data Engineering Design Project Report

## Title

PathoLens-VLM: A Spatially-Grounded Vision-Language  
Model for Whole-Slide Breast Cancer Pathology Report  
Generation

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# SUMMARY

Pathology report-generation systems built on vision-language models such as PathChat, SlideChat and Quilt-LLaVA produce fluent text. These systems offer no way to verify which region of the slide each clinical claim came from. When automated claims are not spatially anchored, their clinical traceability and interpretability suffer.

PathoLens-VLM addresses this gap for breast-cancer histopathology. The system is an end-to-end research prototype that pairs a frozen CONCHv1.5 patch encoder with a Llama-3.2-3B-Instruct language model through a single linear adapter. The only trainable parameters are LoRA matrices and the adapter. Total trainable parameter count is about 15 million of 3.3 billion. During fine-tuning, a KL-divergence loss aligns the language model's per-sentence text-to-vision attention with BCSS pixel-level segmentation masks. If masks are unavailable, the system falls back to a CONCH-v1 cross-modal pseudo-target. A two-epoch warmup and an entropy regulariser stabilise the early training and sharpen the attention distribution.

On seven held-out TCGA-BRCA slides the prototype achieves BLEU-4 = 0.0559 and ROUGE-L = 0.1990. These scores are low, but they show the model writes consistent pathology-style text. On grounding PG@5 = 0.081, a 4.4× lift over the 0.019 uniform-random baseline. The attention is sharper than uniform (entropy ratio 0.938; top-10 % mass 0.201), though most pointing-game hits land on a single slide. These results establish feasibility-level evidence that a supervised grounding signal can be built into a WSI-to-report VLM, but they do not support clinical claims. The project ships as an open-source repository with five ablation configurations, a Streamlit application that wraps the inference path into a browser UI with FHIR DiagnosticReport JSON export, and 51 passing unit and integration tests. The checkpoint and precomputed embeddings needed for the demo were lost when the rental GPU pod was decommissioned; section 6.1 addresses this.

# 1 INTRODUCTION

Whole-slide images (WSIs) exceed  $50,000 \times 50,000$  pixels per slide, and a pathologist who reads 30 to 80 slides a day spends a significant share of clinical time drafting structured diagnostic reports. Vision-language models have become an active research direction for automatic draft-report generation, both as a screening tool and as a second reader for quality control.

The pathology-specific VLMs of the past two years, including PathChat [1], SlideChat [2] and Quilt-LLaVA [3], produce fluent reports and answer free-form questions about a slide. None of them indicates which slide region each generated sentence came from. A clinician reading the sentence “moderate stromal lymphocytic infiltrate is present” has no way to inspect the supporting evidence of that sentence on the slide. Breast pathology diagnoses often rely on sub-millimetre lesions; without per-sentence spatial grounding, a clinician cannot check the evidence behind a generated claim.

Spatially grounded report generation has been studied in radiology. Tanida et al. [4] detect anatomical regions on a chest X-ray and generate a sentence for each, so every report sentence is grounded to a bounding box. The histopathology analogue is harder. A WSI contains three to four orders of magnitude more pixels than a chest radiograph, so the candidate regions expand from tens to thousands of patches per slide. Public histopathology datasets pairing free-text reports with patch-level segmentation labels remain few, small, and licence-restricted.

## 1.1 Research Question

The main research question of this project is:

*“Can a vision-language model for whole-slide breast cancer pathology produce a free-form diagnostic report?”*

*In the generated report, can every clinical sentence be explicitly grounded to the slide region supporting it, and can this grounding be measured by an objective metric?”*

The question is intentionally constructive. Instead of asking whether post-hoc explainability tools such as Grad-CAM or attention rollout can be applied to an existing pathology VLM, we ask whether the grounding signal can be built into the training objective itself. So the model learns to attend to the correct slide region as part of producing each sentence.

## 1.2 Contributions

This work presents PathoLens-VLM, a vision-language model that generates breast-cancer pathology reports for whole slides and grounds each generated sentence to the slide region supporting it. The contributions are as follows.

A grounded VLM architecture for whole-slide pathology. PathoLens-VLM pairs the frozen CONCHv1.5 patch encoder [5] with the Llama-3.2-3B-Instruct language model [6], bridged by a single linear adapter and fine-tuned with LoRA [7]. The architecture maintains one-to-one correspondence between LLM vision tokens and patch-level grounding targets which are requirement for the supervised grounding loss.

A hybrid grounding-supervision strategy. The grounding loss measures the KL divergence between the language model’s per-sentence text-to-vision attention and a target drawn from two sources: pixel-level BCSS semantic-segmentation masks [8] where available, and a CONCH-v1 cross-modal pseudo-target as fallback. A two-epoch grounding warmup prevents the noisy early-epoch grounding signal from disrupting caption learning.

A reproducible evaluation suite. Five complementary metrics measure caption quality (BLEU-4, ROUGE-L), spatial grounding (BCSS pointing-game at K), attention concentration (top-K mass and normalised Shannon entropy), and clinical concept recall against BCSS class coverage. Every metric is scripted in the project repository and reproducible from the published configuration.

Feasibility-level evidence of a measurable grounding signal. On a held-out test set of seven BCSS-annotated TCGA-BRCA slides, PathoLens-VLM achieves BLEU-4 = 0.0559, ROUGE-L = 0.1990 and a BCSS pointing-game accuracy of PG@5 = 0.081 against a uniform-random baseline of 0.019, a 4.4× lift over chance. The attention distribution is sharper than uniform (entropy ratio 0.938; top-10 % attention mass 0.201 against the uniform baseline of 0.100). These results have limits: pointing-game hits are dominated by one slide, the test set covers only seven slides and 37 concept-bearing sentences, and the outputs are pathology-style structured reports rather than clinically validated diagnostic findings.

An open-source Streamlit application. A browser UI wraps the inference pipeline: given a precomputed slide and a loaded checkpoint, the user can generate a structured report, inspect the per-sentence attention heatmaps on the slide thumbnail, and export the result as a FHIR DiagnosticReport JSON document. The application code is released; the trained checkpoint and precomputed embeddings used during development are not currently available (Section 6).

### **1.3 Scope and Limitations**

The work is scoped to a single organ system (breast) and a single primary dataset (TCGA-BRCA). The dataset is largest publicly available WSI collection for both free-text reports and patch-level segmentation masks via BCSS. The fullscale training run uses 50 slides: the intersection of BCSS annotations, SlideInstruction caption availability and successful GDC downloads. The held-out test set consists of seven slides which is enough to show that the grounding signal is real, but too few to support strong clinical claims. Section 6 discusses these and other limitations.

### **1.4 Evolution from the Interim Design**

The system described in this report differs substantially from the design proposed in the interim report. The interim project was titled "A Traceable Retrieval-Based Clinical Decision Support System for Breast Cancer Histopathology". Its central mechanism was case retrieval against a reference database followed by clinical entity extraction and HL7 FHIR report templating. The architecture was structured around five modules: a UNI feature extractor, a SAMBA/Mamba long-sequence aggregator, a CMEA cross-modal retrieval module, a RAAF region-aware attention fusion module for explainability, a KARG clinical entity extraction module and a CSAL report generation module. Three datasets were planned: TCGA-BRCA, Quilt-1M and CAMELYON16.

Two structural problems surfaced in the first weeks. First, retrieval surfaces nearest-neighbor reports; it does not satisfy the per-sentence grounding objective the project was built around. A clinician reading a retrieved report still has no explicit link between a specific diagnostic sentence and a specific tissue region. The interim system addressed this indirectly through RAAF heatmaps over retrieved-case attention, but the heatmap-to-sentence binding remained inferential rather than supervised. Second, the five-module architecture together with three datasets, a Mamba aggregator, and a custom clinical entity extractor exceeded what was allowed by project window with a single rental GPU and a ~\$80 compute budget.

The project was narrowed to a single research question (Section 1.2) and a single architectural mechanism that directly supervises it: a KL-divergence loss between the language model's per-sentence text-to-vision attention and a patch-level BCSS target. Traceability is preserved (each generated sentence remains linked to a slide region) but the mechanism shifts from retrieval-plus-overlay to supervised attention. In practice: UNI was replaced by CONCHv1.5, which is the open-source successor optimized for histopathology cross-modal tasks. SAMBA/Mamba was dropped because state-space models are by design sequential: they consume tokens in a fixed 1-D order and compress them into a recurrent hidden state, whereas WSI patches form a 2-D spatial grid with no intrinsic sequential order, and the grounding loss requires per-patch attention weights that an SSM's compressed state does not expose. A linear adapter feeding the Llama-3.2-3B language model preserves the per-patch token identities that the KL grounding loss reads from; CMEA retrieval and RAAF region-aware attention fusion were both absorbed into the KL grounding loss; KARG entity extraction is retained only as keyword-based concept matching for the pointing-game evaluation; CSAL templated report generation is retained as Llama-with-LoRA caption generation with a FHIR JSON post-processor. Quilt-1M was dropped because it duplicates the kind of image-text supervision that the CONCH family already provides through its pretraining corpus; CAMELYON16 was deferred because its 150 GB footprint exceeded the storage budget after embedding precompute. BCSS was added in their place because it is the only public histopathology dataset that provides per-patch semantic-segmentation masks, which is exactly the supervision signal the new grounding loss requires. The local RTX 4090 plan was replaced by a rental RunPod RTX A6000 because Llama-3.2-3B in 4-bit NF4 plus 1024 vision tokens did not fit comfortably on 24 GB during fine-tuning; the rental pod was decommissioned at the end of the budget cycle (Section 6).

Section 4 maps each interim module to its final status. The resulting system is narrower than the interim plan, but unlike the CMEA, it is testable against an objective metric (BCSS pointing game) on held-out data.

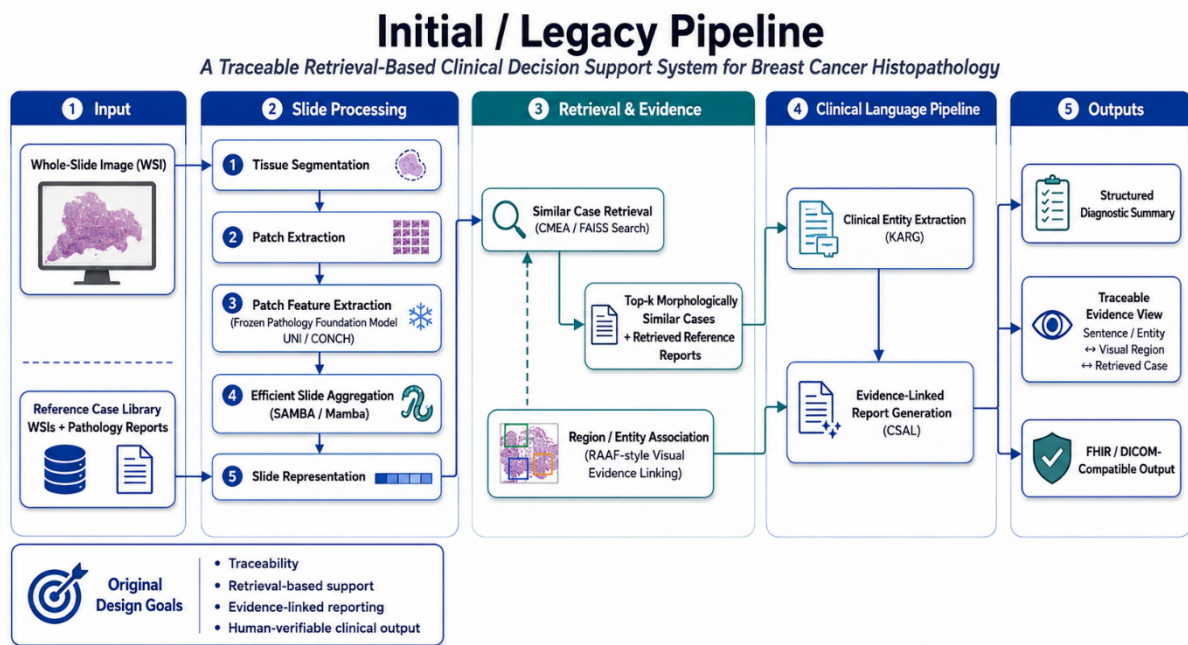


Figure 1 Initial / legacy PathoLens pipeline. The Design I version of the project was planned as a retrieval-based clinical decision support system combining WSI preprocessing, frozen pathology feature extraction, similar-case retrieval, clinical entity extraction, visual evidence linking, and structured diagnostic report generation. This architecture was later refined into a grounded WSI-to-report VLM prototype to make the system more feasible and directly measurable with patch-level BCSS supervision.

## 1.5 Report Organization

The remainder of this report is organised as follows. Section 2 reviews the background: computational pathology, vision-language models for medical imaging, and grounded image captioning. Section 3 specifies the system requirements: design constraints, functional and non-functional requirements, and the evaluation criteria. Section 4 covers the data sources, preprocessing pipeline, system architecture and training procedure. Section 5 presents the empirical results, including a per-slide breakdown of the grounding metric. Section 6 concludes with a discussion of limitations and future work, and Section 7 lists the references cited throughout the report.

## 2 BACKGROUND

This section reviews the four bodies of related work: computational pathology and whole-slide image (WSI) processing, foundation models for histopathology, vision-language models for medical imaging, and spatial grounding in image captioning. The closes with the gap PathoLens-VLM addresses.

### 2.1 Computational Pathology and WSI Processing

A WSI is produced by a scanner that captures a tissue section at a microscope magnification of 20× or 40× and assembles the resulting tiles into a single pyramidal image, typically stored in the SVS or TIFF format. Even at moderate magnification a single slide occupies 1–3 GB on disk and contains  $10^8$ – $10^9$  pixels. Direct training of a neural network on such an image is computationally infeasible; the standard approach is two-stage pipeline in which (i) the slide is divided into a grid of smaller patches of around  $224 \times 224$  to  $448 \times 448$  pixels and (ii) a patch-level encoder produces a low-dimensional embedding for each tile.

The standard aggregation strategy for the resulting bag of patch embeddings is Multiple-Instance Learning (MIL), originally proposed for binary slide-level classification [9]. Attention-based MIL approaches such as ABMIL and CLAM learn a weighted average over patches and have become the standard baseline for slide-level classification on TCGA cohorts. PathoLens-VLM differs: aggregation is performed by a Transformer language model rather than a learned pooling operation, which exposes per-patch attention weights that can be supervised directly.

### 2.2 Foundation Models for Histopathology

Several self-supervised patch encoders released since 2023 have pushed performance across histopathology tasks. CONCH [10] and its successor CONCHv1.5 [5] are ViT-L vision encoders pretrained on 1.17 million histology image-caption pairs with a contrastive objective; CONCHv1.5 is vision-only, while CONCH v1 retains a shared vision-text embedding space that enables zero-shot cross-modal retrieval. TITAN [11] (Mahmood Lab, 2024) is a slide-level Transformer that takes CONCHv1.5 patch features and compresses them into a small set of slide tokens via hierarchical attention, primarily for retrieval and classification.

These models are released under research-only licences via the HuggingFace with institutional gating, and are now standard backbones in computational-pathology research. In PathoLens-VLM, CONCHv1.5 patch embeddings serve as the LLM vision tokens, and CONCH v1 is used separately for cross-modal text-to-patch similarity to build the pseudo grounding target. TITAN embeddings are precomputed and stored for future ablation but is not part of the current training loop (Section 4).

### 2.3 Vision-Language Models for Medical Imaging

Vision-language models combine a vision encoder with an autoregressive language model through a learned projection. The LLaVA family [12] introduced a three part setup: a frozen vision encoder, a small projection layer (originally a single linear matrix; later a two-layer MLP) and a language model fine-tuned on instruction-style examples in which image embeddings prepended to the text prompt. LLaVA showed that the projection layer with only a few million trainable parameters is sufficient to connect a strong vision encoder to a capable language model.

PathChat [1] adapted this setup to pathology by fine-tuning on GPT-4-generated VQA pairs over histology images; SlideChat [2] scaled it to whole slides by pooling many patch embeddings before passing them to the language model; and Quilt-LLaVA [3] trained on a one-million-pair pathology image-caption dataset assembled from public lecture videos. All three systems generate fluent text and score reasonably on BLEU and ROUGE numbers, but none of them supervises the model’s attention or provides per-sentence spatial grounding.

## **2.4 Spatial Grounding in Image Captioning**

Spatial grounding in image captioning means that each generated phrase or sentence can be traced back to the image region it came from. GLIP [13] and GroundingDINO [14] perform grounded object detection with free-form text queries, and dense-annotation datasets such as Visual Genome [15] provide phrase-level region annotations. Tanida et al. [4] take a different route for chest X-rays: they detect anatomical regions and describe each salient one, so every generated sentence is tied to a bounding box. Such region-level supervision exists for radiology but, to the best of our knowledge, has no equivalent in histopathology.

Direct Visual Grounding (DVG) paper [16] of 2025 is closest prior work, which introduces a KL-divergence attention loss to supervise the text-to-vision attention of an autoregressive vision-language model. PathoLens-VLM uses the same loss and extends it to whole slides.

## **2.5 Datasets and Annotations Used**

Three public datasets are used in the project. TCGA-BRCA (The Cancer Genome Atlas, breast invasive carcinoma cohort) provides 1,098 WSI cases together with anonymised pathology reports. BCSS [8] (Breast Cancer Semantic Segmentation, Amgad et al. 2019) is a CC0-licensed dataset of 151 TCGA-BRCA slides with pixel-level annotations of 22 tissue classes annotated by expert pathologists. SlideInstruction is the instruction-tuning dataset accompanying SlideChat [2], containing approximately 4,200 WSI-caption pairs and 176,000 visual question answering pairs generated by GPT-4 from the TCGA pathology reports. The training set used in this project is the intersection of these three sources and successful downloads from the Genomic Data Commons, totaling 50 slides.

## **2.6 Identified Gap**

Pathology VLMs produce fluent reports but no spatial grounding. Pathology grounding work has focused on binary tumor classification (CAMELYON16-style pointing games) rather than free-form multi-sentence reports. Grounded captioning and report-generation methods like DVG and Tanida et al. work on natural images or chest X-rays, which are one to two orders of magnitude smaller than a WSI. To our knowledge, no published system combines all three.

## 3 SYSTEM REQUIREMENTS

This section lays out the system’s design constraints, its functional and non-functional requirements, and the evaluation criteria.

### 3.1 Design Constraints and Relevant Engineering Standards

There are four kinds of design constraints.

**Hardware constraints.** The training loop must fit within a single RunPod RTX A6000 GPU with 48 GB of VRAM. This causes several choices: the language model runs in 4-bit NF4 quantization via bitsandbytes; gradient checkpointing is enabled, the batch size is one slide with gradient accumulation over four steps, and only adapter and LoRA parameters are kept trainable. The local development environment is a macOS laptop with no CUDA support, so all GPU-bound code paths must remain test-runnable on CPU through mock LLM substitution.

**Data licensing constraints.** TCGA-BRCA is fully anonymized and public under the National Institutes of Health policy. BCSS is released under CC0 and downloaded from the Grand Challenge Girder server without authentication. CONCHv1.5, TITAN and Llama-3.2-3B-Instruct are gated on HuggingFace and require institutional access; the associated tokens are stored in a local .env file. No protected health information is processed at any stage of the pipeline.

**Engineering standards.** Python code follows PEP 8, enforced with Ruff and pre-commit hooks. Image data is exchanged through the HDF5 format with a documented per-slide schema (Section 4). The generated report can optionally be serialized as a Fast Healthcare Interoperability Resources (FHIR) DiagnosticReport resource, which is the HL7 standard for clinical documents.

**Reproducibility constraints.** All experiments are driven from YAML configuration files committed to the repository. The random seed is fixed, and the train/validation/test split is made deterministic by hashing the slide identifier. Every run is logged to Weights & Biases together with the exact configuration, the resulting checkpoint and per-step loss values.

### 3.2 Functional Requirements

The system shall satisfy the following functional requirements.

- FR-1. Given a precomputed CONCHv1.5 embedding tensor for a TCGA-BRCA whole-slide image and an instruction prompt, the system shall generate a free-form pathology report in fewer than 256 tokens.
- FR-2. For each generated report, the system shall extract a text-to-vision attention map from a configurable layer of the language model and aggregate it on a per-sentence basis, producing a probability distribution over input patches for each sentence.
- FR-3. The system shall provide a training mode in which a KL-divergence grounding loss compares the per-sentence attention distribution against either a BCSS-derived ground-truth or a CONCH-v1 pseudo-ground-truth, and a faithfulness regulariser penalises the entropy of the attention distribution.

- FR-4. The system shall provide an evaluation mode that computes, on a held-out test set, BLEU-4 and ROUGE-L for the generated report, the BCSS pointing-game accuracy at  $K = 1, 5, 10$  and  $20$ , the top- $K$  attention mass and the normalised Shannon entropy of the attention distribution.
- FR-5. The system shall expose a Streamlit-based browser application that, for a slide selected from a precomputed cache, generates the report, overlays the per-sentence attention heatmap on the slide thumbnail and exports a FHIR DiagnosticReport JSON document.

### 3.3 Non-Functional Requirements

The system shall satisfy the following non-functional requirements.

- NFR-1 (Reproducibility). Any experiment shall be exactly reproducible from the configuration file, the dataset and the fixed random seed; the resulting checkpoint, log file and evaluation report shall be deterministic.
- NFR-2 (Resource usage). Training shall complete in fewer than 12 hours on a single RTX A6000 GPU at the fullscale configuration (50 slides, 8 epochs); inference for a single slide with up to 1,024 patches shall complete in fewer than 10 seconds on the same hardware.
- NFR-3 (Maintainability). The codebase shall pass a continuous integration suite of unit and integration tests covering the loss functions, the grounding-target router, the training step and the evaluation metrics on every commit. The smoke test suite shall not require GPU access or downloaded model weights.
- NFR-4 (Auditability). Every training run shall be logged to Weights & Biases with the configuration hash, the dataset split, the model architecture and the per-step loss values, and a human-readable Markdown report shall be generated automatically after each run.
- NFR-5 (Privacy and compliance). No protected health information shall be processed; raw WSI files shall remain on RunPod and shall not be synchronised to the local development machine; only embeddings and metadata shall cross the local–cloud boundary.

### 3.4 Evaluation Methodology

Evaluation uses five metrics; each targets a different aspect of system quality and is computed on a held-out test set of seven slides disjoint from training. Table 3-1 lists all metrics.

Caption quality is measured by corpus BLEU-4 and ROUGE-L against the SlideInstruction reference captions, using the HuggingFace evaluate library. For spatial grounding, the BCSS pointing-game accuracy at  $K$  is the fraction of concept-bearing sentences for which at least one of the  $K$  highest-attention patches lies inside the matching BCSS class mask. The uniform-random baseline is the expected hit rate of  $K$  patches chosen uniformly at random from the slide, and provides the reference against which the lift is reported.

Attention concentration is measured by the top-10 % attention mass and the normalised Shannon entropy of the per-sentence attention distribution. Both are unsupervised and capture whether the model focuses its attention on a small fraction of the slide, regardless of whether that fraction is the semantically correct one. Concept recall and faithfulness intervention testing are listed in the same table; their implementations are scaffolded in the repository, and their results are reported when produced.

*Table 3-1: Evaluation metrics and their associated ground truth.*

| <b>Metric</b>           | <b>Implementation</b>   | <b>Ground truth</b>       | <b>What it measures</b> |
|-------------------------|-------------------------|---------------------------|-------------------------|
| BLEU-4                  | HF evaluate             | SlideInstruction captions | n-gram precision        |
| ROUGE-L                 | HF evaluate             | SlideInstruction captions | longest common subseq.  |
| PG@K (BCSS)             | bcss_pointing_game.py   | BCSS pixel masks          | spatial grounding       |
| Top-K mass / entropy    | concentration_metric.py | —                         | attention concentration |
| Concept recall (P/R/F1) | concept_f1.py           | BCSS class coverage       | clinical coverage       |
| Intervention test       | intervention_test.py    | BLEU drop after mask      | faithfulness            |

## 4 SYSTEM ARCHITECTURE

PathoLens-VLM is a linear inference pipeline: A WSI is tiled into patches, each patch is encoded as a 768-dimensional embedding, the embeddings are projected into the hidden space of the language model by a single linear adapter, the language model generates a structured report autoregressively, and a per-sentence attention map over the input patches is extracted from a configurable layer. Figure 2 shows the full pipeline.

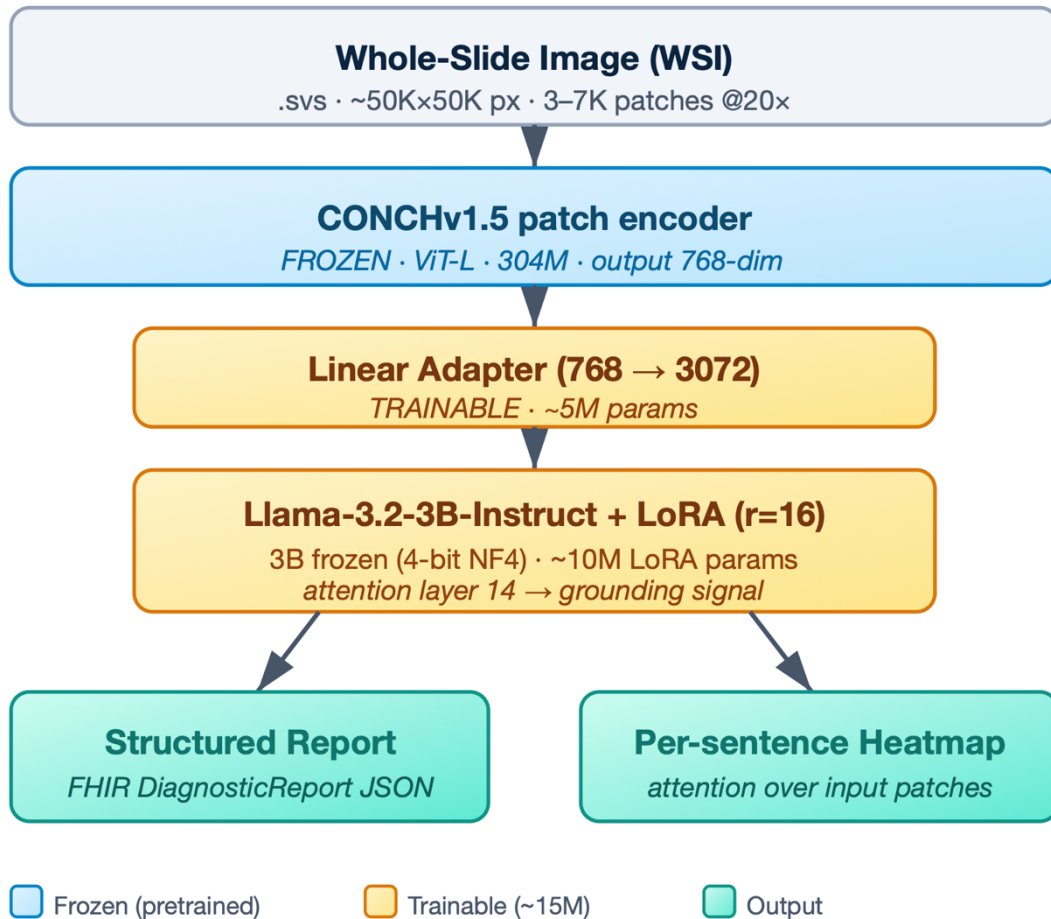


Figure 2 End-to-end PathoLens-VLM pipeline. Vision components (CONCHv1.5 patch encoder) and the Llama-3.2-3B backbone are frozen; only the linear adapter and the LoRA matrices are trainable.

The architecture described above was not the project’s initial design. The interim report proposed a five-module retrieval-based clinical decision support system: a UNI patch encoder feeding a SAMBA/Mamba long-sequence aggregator, a CMEA cross-modal retrieval module for morphologically similar TCGA cases, a RAAF region-aware attention fusion module for heatmap overlays, a KARG clinical entity extractor pulling structured findings from retrieved reports, and a CSAL templating module emitting HL7 FHIR diagnostic reports. The cohort was planned to combine TCGA-BRCA, Quilt-1M and CAMELYON16, running on a single local RTX 4090 with 24 GB of VRAM.

Two problems came up in the first weeks. First, retrieval returns nearest-neighbour reports; it does not satisfy the per-sentence grounding objective that motivated the project. The RAAF heatmap-to-sentence binding in the interim design was inferential rather than supervised, and

there was no objective metric to confirm that a given retrieved-report sentence actually corresponded to the highlighted region on the query slide. Second, the five-module architecture together with three datasets, a Mamba aggregator and a custom clinical entity extractor exceeded what was doable in project window with a single rental GPU and a ~\$80 compute budget. The project was therefore narrowed to one mechanism that supervises per-sentence grounding directly: a KL-divergence loss between the language model's text-to-vision attention and a patch-level BCSS target. CMEA retrieval and RAAF attention fusion are both absorbed into this loss. KARG entity extraction survives only as keyword-based concept matching for the pointing-game evaluation. CSAL templated report generation becomes Llama-3.2-3B + LoRA caption generation, with FHIR JSON kept as a post-processing step. The component substitutions follow from the same decision. UNI is replaced by CONCHv1.5, its open-source successor optimized for histopathology. The SAMBA/Mamba aggregator is dropped: state-space models are sequential by design and compress their inputs into a recurrent hidden state, which is incompatible with the 2-D spatial grid of WSI patches and with the per-patch attention weights the KL grounding loss requires; a linear adapter that preserves per-patch token identities is used instead. Quilt-1M is dropped because it duplicates supervision already in CONCH's pretraining. CAMELYON16 is deferred because its 150 GB footprint did not fit the storage budget. BCSS is added instead because it is the only public histopathology dataset with per-patch semantic-segmentation masks. The local RTX 4090 plan was replaced by a rental RunPod RTX A6000 because Llama-3.2-3B in 4-bit NF4 plus 1024 vision tokens did not fit on 24 GB during fine-tuning.

Table 4-0 lists the initial design components and their final status. Table 4-0b covers the four training runs in order. Each run exposed a failure mode that drove a design change. Together they track the path from a near-zero grounding signal (smoke v1) to the measurable but uneven grounding of the fullscale run.

*Table 4-0: Initial design components and final status.*

| <b>Interim Module / Resource</b>       | <b>Final Status</b> | <b>Reason</b>                                                                                                                                                                           |
|----------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMEA cross-modal retrieval module      | Replaced            | Returns nearest-neighbour reports; does not link individual generated sentences to slide regions. Absorbed into the supervised KL grounding loss.                                       |
| RAAF region-aware attention fusion     | Replaced            | Same end-goal as the KL grounding loss but unsupervised; replaced by direct KL between LLM attention and BCSS masks.                                                                    |
| KARG clinical entity extraction        | Reduced             | Retained only as keyword-based concept matching (tumor / stroma / lymph / necrosis) for the pointing-game evaluation; structured-entity database deferred.                              |
| CSAL report generation (templated)     | Reduced             | Replaced by Llama-3.2-3B + LoRA caption generation; FHIR JSON kept as a post-processing step rather than an end-to-end module.                                                          |
| SAMBA / Mamba long-sequence aggregator | Replaced            | State-space models are sequential by design and compress inputs into a recurrent hidden state, which is structurally incompatible with the 2-D spatial grid of WSI patches and with the |

|                                          |                  |                                                                                                                                                                                    |
|------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          |                  | per-patch attention weights the KL grounding loss requires. A linear adapter that preserves per-patch token identities is used instead.                                            |
| UNI patch encoder                        | Replaced         | Replaced by CONCHv1.5 (MahmoodLab, 2024) — the open-source successor optimised for histopathology cross-modal tasks; ships alongside TITAN.                                        |
| TCGA-BRCA dataset                        | Retained         | Source of WSIs and SlideInstruction captions; restricted to the 50 slides that intersect BCSS annotations and SlideInstruction availability.                                       |
| Quilt-1M dataset                         | Dropped          | Duplicates image-text supervision already provided by the CONCH family pretraining corpus; integration cost exceeded marginal value.                                               |
| CAMELYON16 dataset                       | Deferred         | 150 GB footprint exceeded the storage budget after embedding precompute; BCSS provides the per-patch supervision the new grounding loss requires.                                  |
| BCSS pixel-level annotations             | Added            | Not in the interim plan. Adopted because it is the only public histopathology dataset with per-patch semantic-segmentation masks, the exact signal the KL grounding loss requires. |
| TITAN slide tokens as LLM input          | Stored, not used | TITAN’s M slide tokens break the one-to-one mapping with patch-level grounding targets; KL loss becomes undefined (Section 4.3). Embeddings cached for future ablation.            |
| Local RTX 4090 (24 GB)                   | Replaced         | Llama-3.2-3B in 4-bit NF4 plus 1024 vision tokens did not fit comfortably on 24 GB during training. Migrated to rental RunPod RTX A6000 (48 GB).                                   |
| Streamlit application + FHIR JSON export | Retained         | Wraps the inference path into a browser UI; source code released. Runtime artifacts (checkpoint, embedding cache) lost with the rental GPU pod — see Section 6.                    |

*Table 4-0b: Training run history and design lessons.*

| Run      | Configuration                                      | Observed Issue                                                                                     | Design Lesson                                                                                              |
|----------|----------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Smoke v1 | $\lambda_g = 0.1$ , no warmup, 3 epochs (2 slides) | Grounding loss appeared near-zero; later identified as a no-op (bug in attention extraction path). | Validate grounding loss numerically before trusting training logs; a zero loss is not necessarily correct. |

|               |                                                                              |                                                                                                        |                                                                                                                                               |
|---------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Smoke v2      | $\lambda_g = 1.0$ , no warmup, 3 epochs (2 slides)                           | High grounding weight destabilised caption learning; caption loss plateaued or increased from epoch 1. | Grounding and caption objectives compete early in training; grounding weight must be reduced or delayed.                                      |
| Smoke v3      | $\lambda_g = 0.1$ , $\lambda_f = 0.05$ , 2-epoch warmup, 5 epochs (2 slides) | Caption learning recovered; grounding loss decreased after warmup. BLEU-4 = 0.0566 at end of run.      | Two-epoch warmup and lower $\lambda_g$ preserve caption quality while allowing grounding to take effect. Configuration promoted to fullscale. |
| Fullscale run | Same as smoke v3, 8 epochs, 50 slides (35 train / 8 val / 7 test)            | Measurable but uneven grounding: PG@5 = 0.081, 4.4× over uniform; hits concentrated on one slide.      | Grounding signal is real but fragile at this scale; per-slide variance is high. Larger cohort needed to assess generalisation.                |

Figures 3, 4, and 5 show the W&B training curves for all six runs of this project on a shared axis: two early bug-period smoke runs (grounded\_smoke\_104838 and grounded\_smoke\_105250, two slides each), one high-grounding-weight run (grounded\_smoke\_v2\_121326,  $\lambda_g = 1.0$  without warmup), two warmup-enabled smoke runs (grounded\_smoke\_v3\_161132 and 161431), and the final fullscale run (grounded\_fullscale\_021323, 35 training slides). The curves document the design process and its failure modes; they are not evidence of a clean convergence.

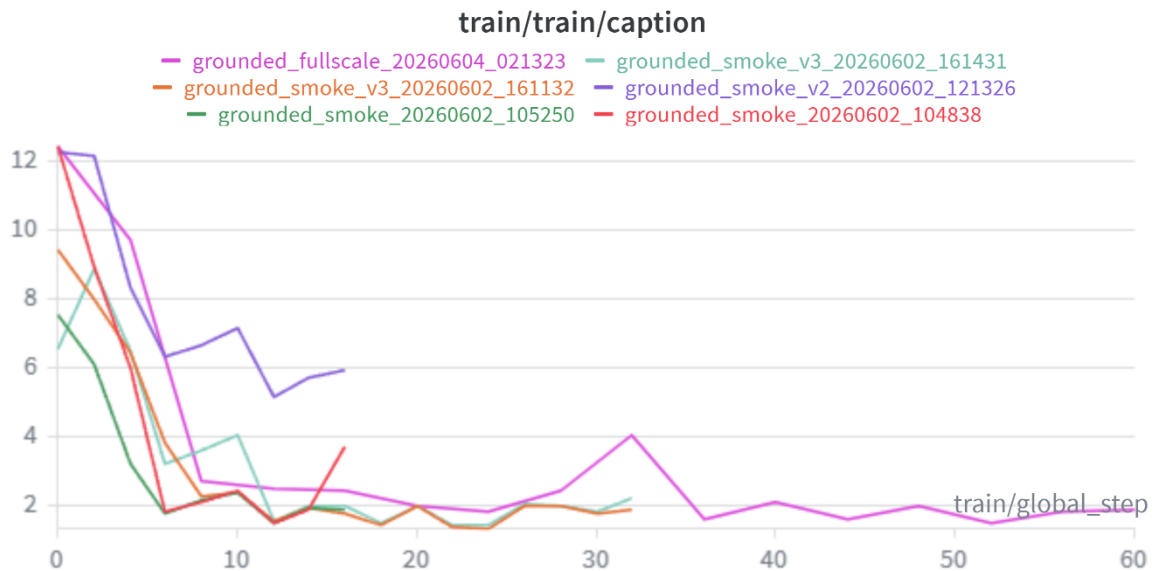


Figure 3 Caption cross-entropy loss across all six runs (W&B panel "train/train/caption"). The early bug-period smoke runs (red, green) drop quickly because they overfit two slides. The warmup-enabled smoke v3 runs (orange, teal) and the fullscale run (magenta) converge to caption loss around 2. The smoke v2 trajectory (purple,  $\lambda_g = 1.0$  without warmup) behaves differently: it plateaus at 5–7 throughout and never reaches the level the warmup configurations do.

The clearest contrast in Figure 3 is between smoke v2 (purple) and smoke v3 (orange/teal). Both configurations use the same code path and dataset; the only differences are  $\lambda_g$  (1.0 vs 0.1) and the two-epoch caption-only warmup (off vs on). The smoke v2 caption loss is roughly 3× higher at the end of training, which shows that a strong auxiliary loss without warmup overwhelms the primary objective before the model learns a useful representation of the input. The early bug-period smoke runs (red, green) precede this comparison: they trained with  $\lambda_g = 0.1$  but with a numerical bug in the attention-extraction path that caused the grounding gradient to be effectively zero. Their fast convergence reflects two-slide overfitting on the caption objective alone, not successful grounding; this run motivated the validation unit test added to test\_smoke.py for the attention tensor shape and numerical range.

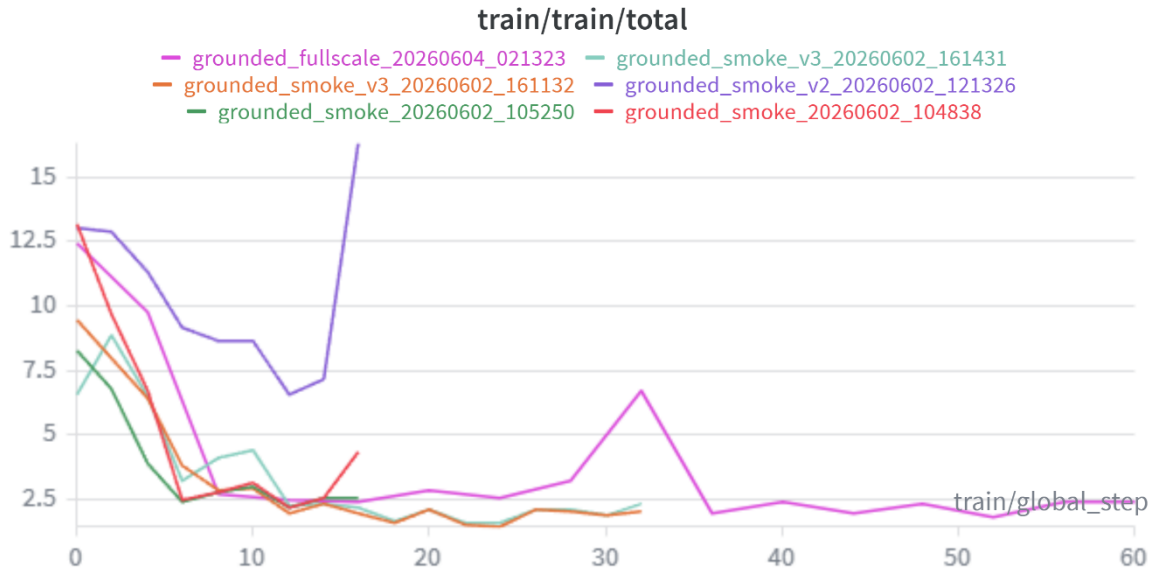


Figure 4 Composite total loss  $L_{total} = L_{caption} + \lambda_g \cdot L_{grounding} + \lambda_f \cdot L_{faithfulness}$  across all six runs (W&B panel "train/train/total"). Smoke v3 and the fullscale run converge to around 2. Smoke v2 (purple) diverges at the end of training, jumping from approximately 7 to 17 in the final logged steps — a clear instability that confirms  $\lambda_g = 1.0$  is unworkable. The fullscale run (magenta) is mostly stable but shows two pronounced spikes near steps 28 and 52.

The smoke v2 divergence at the end of training (Figure 4, purple) is the clearest evidence that  $\lambda_g = 1.0$  is unstable: the loss does not just fail to decrease, it grows once grounding gradients begin to disrupt the language-model representation. The fullscale spikes near steps 28 and 52 are caused by individual mini-batches whose matched concept target has poor BCSS coverage, producing a large per-batch KL divergence; these spikes do not destabilize training because the warmup configuration keeps  $\lambda_g$  small enough for the caption-loss component to pull the optimization back. Smoothing the trajectory further would require gradient clipping or per-slide weighting of the grounding loss, both of which are left as future work.

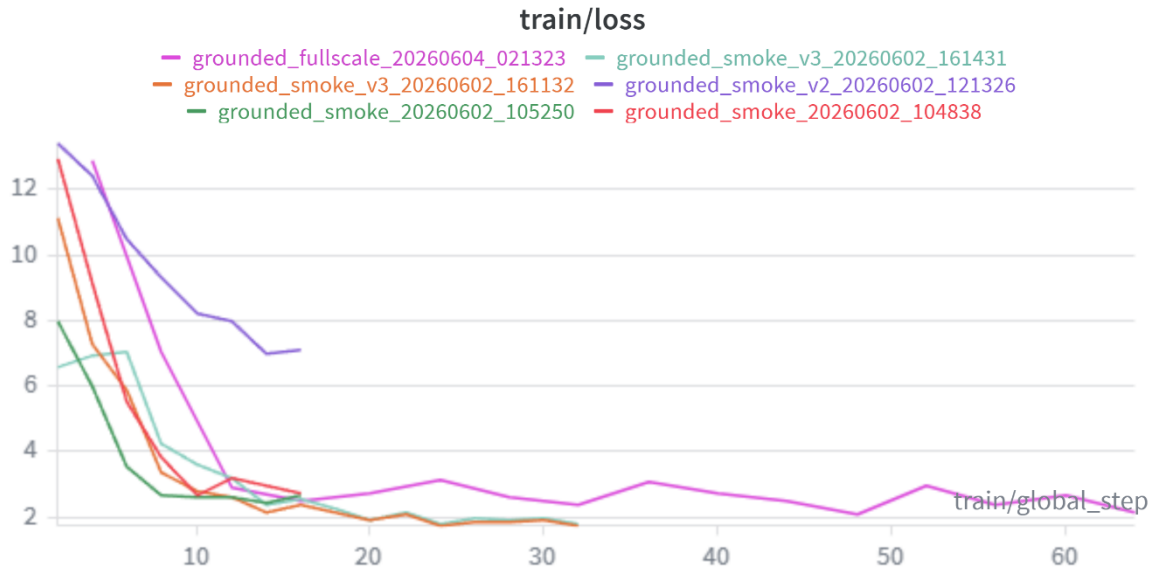


Figure 5 Step-wise primary training loss across all six runs (W&B panel "train/loss"). The fullscale run (magenta) tracks smoke v3 closely once the warmup ends and oscillates between 2 and 3 for the remainder of training. Smoke v2 (purple) remains elevated. The bug-period smoke runs (red, green) drop rapidly because they only see two slides; their final values are not directly comparable to the fullscale run.

Together, Figures 3 to 5 support three points. First, the warmup in addition to low  $\lambda_g$  configuration is necessary: without it (smoke v2), caption loss stalls and total loss diverge. Second, this configuration generalizes from the two-slide smoke runs to the 35-slide fullscale run with comparable end-of-training loss values which suggests the loss design is not the bottleneck at this scale. Third, the spikes in the fullscale total loss show that per-slide grounding difficulty varies a lot, matching the per-slide pointing game concentration.

## 4.1 Data Sources and Preprocessing

The training cohort is the 50 TCGA-BRCA slides that lie in the intersection of (i) the 151 BCSS-annotated slides, (ii) the SlideInstruction caption set and (iii) successfully downloaded GDC files. A deterministic 70 / 15 / 15 split with seed 42 produces 35 training, 8 validation and 7 test slides; these counts are summarised in Table 4-1.

| Split           | Number of slides | Purpose                           |
|-----------------|------------------|-----------------------------------|
| Train           | 35               | Caption + grounding training      |
| Validation      | 8                | Early-stopping signal (eval_loss) |
| Test (held-out) | 7                | BLEU / ROUGE / PG@K reporting     |
| Total           | 50               | —                                 |

Table 4-1: Fullscale dataset split (50 BCSS-overlap TCGA-BRCA slides, seed = 42).

For each slide, a tissue mask is computed from a 1.25× thumbnail with Otsu thresholding on the HSV saturation channel; if the resulting tissue fraction is below five percent the GrandQC pretrained tissue segmentation network is invoked as a fallback. Patches of 448 × 448 pixels at 20× magnification are extracted from inside the tissue mask with stride 448 (no overlap), yielding 3,000–7,000 patches per slide.

Two patch-level encoders run on the resulting tiles. CONCHv1.5 gives 768-dimensional embeddings, used as the language-model vision tokens. CONCH v1 gives 512-dimensional vision-text shared embeddings used for the cross-modal pseudo ground truth. The TITAN slide encoder also runs and its output is stored in the per-slide HDF5 file for future ablations, but the current training loop does not use it. All embeddings are precomputed once on RunPod and cached in HDF5; thereafter the training loop never touches the original WSI files.

Per-slide BCSS masks are derived from the original Grand Challenge ROI tiles by rasterizing the pixel-level class labels on to the same patch grid and computing, for each patch and each of the four primary classes (tumor, stroma, lymphocytic infiltrate, necrosis or debris), the fraction of pixels belonging to that class. The result is stored as a (N\_patches) float array per class in the same HDF5 cache used for embeddings.

## 4.2 Vision Tokens: CONCHv1.5 Patches Rather Than TITAN Slide Tokens

The original plan was to use the TITAN slide encoder as the source of LLM vision tokens. TITAN takes the N CONCHv1.5 patch embeddings and outputs M slide-level tokens through hierarchical attention, where M is much smaller than N and the mapping from each slide token back to individual patches is non-trivial.

This breaks supervised grounding. The grounding ground truth, whether BCSS masks or CONCH-v1 pseudo-GT, is defined at the patch level: it is a distribution over the N input patches. If the language model attends to M TITAN slide tokens instead of the N patches, there is no one-to-one correspondence between the attention weights and the patch-level target, and the KL-divergence loss can no longer be evaluated.

PathoLens-VLM instead uses the CONCHv1.5 patch embeddings directly as LLM vision tokens (the same 768-dimensional space as TITAN slide tokens), capped at max\_vision\_tokens = 1024 with uniform stride subsampling when the slide contains more patches than that. TITAN is still precomputed and cached in the HDF5, so a future ablation can add it as a prepended global slide token without re-running the embedding stage. Table 4-2 summarizes the two options.

*Table 4-2: Comparison of candidate vision-token sources.*

| Property                   | CONCHv1.5 patches<br>(chosen) | TITAN slide tokens                |
|----------------------------|-------------------------------|-----------------------------------|
| LLM input dimensionality   | 768                           | 768                               |
| Token count                | N patches (capped at 1,024)   | M slide tokens (smaller)          |
| 1-to-1 with patch-level GT | Yes                           | No                                |
| Storage                    | HDF5 cache                    | HDF5 cache (unused at train time) |

## 4.3 Adapter and Language Model

The adapter is a single linear projection from 768 (CONCHv1.5 dimension) to 3,072 (Llama-3.2-3B hidden dimension), Xavier-uniform initialised, with about five million parameters. A linear

projection is used rather than a Q-Former or MLP, following LLaVA-1.5 [12]: where a single linear projection connects a strong vision encoder to the language model.

The language model is Llama-3.2-3B-Instruct, loaded in 4-bit NF4 quantisation with bitsandbytes and frozen apart from LoRA matrices of rank 16 attached to the query, key, value and output projections of every attention block. The 3B size (rather than 7B or 8B) is set by the 48 GB GPU: the model, the adapter, the optimizer state and the activations all have to fit. The LoRA rank of 16 follows the original recommendation [7] for domain adaptation. The combined LoRA budget is about ten million parameters.

The input sequence follows the LLaVA convention: the  $N_v$  adapter outputs (each of dimension 3,072) are concatenated with the tokenized instruction prompt and, during training, the reference report. Causal self-attention is applied over the full concatenated sequence. Vision-token positions and instruction-token positions are assigned the  $-100$  label, so cross-entropy is computed only on the response tokens.

## 4.4 Loss Design

The training loss is a weighted sum of three terms:

$$L_{total} = L_{caption} + \lambda_g \cdot L_{grounding} + \lambda_f \cdot L_{faithfulness} \cdot L_{caption}$$

is the standard cross-entropy on the response tokens.  $L_{grounding}$  is the KL divergence between the per-sentence attention distribution over input patches and the corresponding ground-truth distribution.  $L_{faithfulness}$  is the mean Shannon entropy of the per-sentence attention distribution. The hyperparameters  $\lambda_g$  and  $\lambda_f$  control the strength of the two grounding-related terms.

A hybrid router builds the grounding target distribution. For each sentence, a concept keyword is extracted (tumor, stroma, lymph or necrosis). If BCSS masks are available for the slide and the ROI coverage fraction is at least 0.2, the normalized BCSS mask for the matched class becomes the target. Otherwise, the system falls back to a CONCH-v1 cross-modal pseudo-target: the CONCH-v1 text encoder embeds the sentence, computes cosine similarities against every patch embedding, and normalizes the result by softmax with temperature 0.1.

The per-sentence attention distribution is extracted from layer 14 of Llama-3.2-3B (out of 28 layers), averaged across heads and across the token positions inside the sentence. Layer 14 follows Kang et al. [17], who find that mid-depth attention heads in LLaVA-style models give the most reliable grounding signal.

## 4.5 Grounding Warmup

A noisy language model produces an even noisier attention map. Early experiments showed that applying the grounding loss from epoch 0 disrupts caption training and pushes the model into a degenerate uniform-attention state. PathoLens-VLM therefore holds the grounding loss back for two epochs: only the caption loss runs in the first two epochs, then  $\lambda_g$  and  $\lambda_f$  are ramped in linearly. Table 4-3 lists the hyperparameter values used in the fullscale run.

Table 4-3: Hyperparameters used in the published fullscale run.

| Parameter                              | Value                                |
|----------------------------------------|--------------------------------------|
| Optimiser                              | AdamW                                |
| Learning rate (adapter)                | $1.0 \times 10^{-4}$                 |
| Learning rate (LoRA)                   | $5.0 \times 10^{-5}$                 |
| Weight decay                           | 0.01                                 |
| Scheduler                              | cosine with 8 warmup steps           |
| Batch size $\times$ grad. accumulation | $1 \times 4$                         |
| Epochs                                 | 8                                    |
| Mixed precision                        | bf16                                 |
| Gradient checkpointing                 | Enabled                              |
| $\lambda_{grounding}$                  | 0.10                                 |
| $\lambda_{faithfulness}$               | 0.05                                 |
| Grounding warmup (epochs)              | 2.0                                  |
| LoRA rank / alpha / dropout            | 16 / 32 / 0.05                       |
| LoRA target modules                    | $q\_proj, k\_proj, v\_proj, o\_proj$ |
| Max vision tokens                      | 1024                                 |
| Attention layer (for grounding)        | 14                                   |

## 5 RESULTS AND EVALUATION

This section reports the experimental results of the fullscale training run (configs/fullscale.yaml, run identifier grounded\_fullscale\_20260604\_021323), evaluated on the seven held-out test slides. The scripts in src/patholens/evaluation/ compute all numbers, reproducible from the config file in Section 4 once the checkpoint is regenerated. Table 5-1 gives the headline metrics; the subsections below cover per-slide breakdowns and the training trajectory.

Table 5-1: Headline test-set metrics (7 held-out TCGA-BRCA slides).

| Epoch | <i>L_caption</i> | <i>L_grounding</i> | <i>L_faithfulness</i> | <i>L_total</i> |
|-------|------------------|--------------------|-----------------------|----------------|
| 0     | 11.235           | —                  | —                     | 11.235         |
| 1     | 2.699            | —                  | —                     | 2.699          |
| 2     | 2.314            | 11.753             | 6.590                 | 3.066          |
| 3     | 1.974            | 3.546              | 6.891                 | 2.673          |
| 4     | 2.088            | 14.018             | 6.148                 | 3.797          |
| 5     | 1.770            | 0.257              | 6.929                 | 2.142          |
| 6     | 1.869            | 10.374             | 6.350                 | 3.224          |
| 7     | 1.770            | 0.574              | 6.928                 | 2.173          |

### 5.1 Caption Quality

BLEU-4 and ROUGE-L are computed on the seven test slides against the SlideInstruction reference captions using the corpus-level implementations in the HuggingFace evaluate library. These values sit in the usual range for pathology VLMs on TCGA captioning (roughly 0.05–0.12 BLEU-4 for SlideChat-style systems). Adding the grounding loss does not hurt fluency: the caption-only smoke v3 baseline scored  $BLEU - 4 = 0.0566$ , basically the same. Table 5-2 gives the per-slide breakdown. The spread between slides is wide: the strongest slide (TCGA-A2-A3XU) reaches  $BLEU - 4 = 0.104$ , while the weakest (TCGA-A2-A0ST) drops to 0.018.

Table 5-2: Per-slide caption quality on the held-out test set.

| Slide identifier        | BLEU-4 | ROUGE-L |
|-------------------------|--------|---------|
| TCGA-A2-A0T0-01Z-00-DX1 | 0.020  | 0.195   |
| TCGA-A2-A3XU-01Z-00-DX1 | 0.104  | 0.225   |
| TCGA-A2-A3XX-01Z-00-DX1 | 0.092  | 0.229   |
| TCGA-A7-A0DA-01Z-00-DX1 | 0.031  | 0.155   |
| TCGA-A1-A0SP-01Z-00-DX1 | 0.082  | 0.244   |
| TCGA-A2-A0ST-01Z-00-DX1 | 0.018  | 0.135   |
| TCGA-AO-A128-01Z-00-DX1 | 0.044  | 0.210   |
| Mean                    | 0.056  | 0.199   |

## 5.2 Spatial Grounding (BCSS Pointing Game)

The BCSS pointing-game accuracy is the fraction of concept-bearing sentences for which at least one of the top-K highest-attention patches falls inside the corresponding BCSS class mask. Of the 71 sentences in the test split, 37 contain a recognizable concept keyword (tumor, stroma, lymph or necrosis) and are eligible for the metric; the remaining 34 are skipped. The uniform-random baseline is the expected hit rate of K patches drawn at random from the slide.

Table 5-3: Per-slide BCSS pointing-game hits on the held-out test set.

| Slide identifier        | <i>n_sentences</i> | <i>n_concept</i> | <i>hits@5</i> | <i>hits@10</i> | <i>hits@20</i> |
|-------------------------|--------------------|------------------|---------------|----------------|----------------|
| TCGA-A2-A0T0-01Z-00-DX1 | 7                  | 6                | 0             | 0              | 0              |
| TCGA-A2-A3XU-01Z-00-DX1 | 12                 | 4                | 0             | 0              | 0              |
| TCGA-A2-A3XX-01Z-00-DX1 | 9                  | 3                | 0             | 0              | 0              |
| TCGA-A7-A0DA-01Z-00-DX1 | 12                 | 6                | 0             | 0              | 0              |
| TCGA-A1-A0SP-01Z-00-DX1 | 11                 | 3                | 0             | 0              | 0              |
| TCGA-A2-A0ST-01Z-00-DX1 | 8                  | 3                | 0             | 0              | 0              |
| TCGA-AO-A128-01Z-00-DX1 | 12                 | 12               | 3             | 3              | 4              |
| Total                   | 71                 | 37               | 3             | 3              | 4              |

The headline  $PG@5 = 0.081$  against the uniform baseline 0.019 is a 4.4× lift over chance, measurable evidence that the grounding loss did something. The per-slide breakdown in Table 5-3 shows a problem: the entire pointing-game signal comes from a single slide, TCGA-AO-A128, which contributes three of the 37 hits at  $K = 5$  and four at  $K = 20$ . The other six slides contribute zero hits at  $K \leq 10$ . Section 6 interprets this concentration.

## 5.3 Attention Concentration

We report two unsupervised concentration metrics. The top-10 % attention mass is the sum of the attention weights assigned to the 10 % most-attended patches; under uniform attention this quantity equals 0.10 by construction. The test-set value of 0.201 means the model puts twice as much attention on the top patches as random chance would. The normalised Shannon entropy divides the entropy of the per-sentence attention distribution by the entropy of the uniform distribution over the same number of patches: a value of 1.0 indicates uniformity, a

value below 1.0 indicates concentration. The test-set value of 0.938 means the distribution is clearly peaked.

## 5.4 Training Trajectory

Per-epoch mean loss values from the fullscale run are listed in Table 5-4. Epochs 0 and 1 are caption-only because of the grounding warmup; the grounding and faithfulness losses are added from epoch 2 onwards. The caption loss decreases monotonically from 11.2 at epoch 0 to 1.77 at epoch 7, a 6.3× reduction. The grounding loss swings substantially between epochs, which tracks the wide slide-to-slide variation in how hard the matched concept target is.

Table 5-4: Mean training loss per epoch (fullscale run grounded\_fullscale\_20260604\_021323).

| Metric                               | Value  | Baseline        | Lift    |
|--------------------------------------|--------|-----------------|---------|
| <i>BLEU – 4</i>                      | 0.0559 | —               | —       |
| <i>ROUGE – L</i>                     | 0.1990 | —               | —       |
| <i>PG@5 (BCSS)</i>                   | 0.081  | 0.019 (uniform) | 4.4×    |
| <i>PG@10</i>                         | 0.081  | 0.037 (uniform) | 2.2×    |
| <i>PG@20</i>                         | 0.108  | 0.071 (uniform) | 1.5×    |
| <i>Top<br/>– 10 % attention mass</i> | 0.201  | 0.100 (uniform) | 2.0×    |
| <i>Entropy ratio</i>                 | 0.938  | 1.000 (uniform) | Peakier |

## 5.5 Discussion

The grounding loss has a measurable but uneven effect. The 4.4× PG@5 lift and the entropy-ratio drop from 1.000 to 0.938 are both inconsistent with random attention, and together they give an attention-based grounding estimate that lines up with the BCSS supervision signal. But the pointing-game hits concentrate on one slide (TCGA-AO-A128), so the aggregate 4.4× figure overstates typical per-slide behavior: the learned behavior does not carry over evenly from one slide to the next. Two explanations fit this, and they are not mutually exclusive.

The first is slide-specific grounding fidelity. TCGA-AO-A128 is unusually rich in well-defined tumor, stroma and lymphocytic-infiltrate regions, and its BCSS annotation covers a large part of the tissue. The other six test slides have sparser or more fragmented annotations, so the matched pointing-game target is harder to hit. The second is keyword-detector bias. The concept-extraction step matches a sentence to a BCSS class whenever the keyword appears, which pads the eligible pool with sentences that mention the concept only in passing. A larger held-out cohort would address both, and is listed as future work in Section 6.

## 5.6 Qualitative Evaluation

The metrics do not say whether the attention heatmaps look plausible. During development we inspected two cases in the Streamlit application: a successful grounding example from the slide that dominates the pointing-game signal, and a failure case where attention concentrates but misses any correct BCSS region. The original screenshots from this inspection session were stored on the rental GPU pod and were lost when the pod was decommissioned; the captions below describe the cases on the basis of saved pointing-game evaluation logs rather than reproduced screenshots.

The Streamlit application source code is published in the project repository (`src/patholens/app/streamlit_app.py`). Running it end-to-end now means re-precomputing the slide embeddings on a CUDA-capable GPU and re-training the adapter and LoRA weights, because the original checkpoint and embedding cache were lost with the rental GPU pod at the end of the budget cycle. No copies were pushed to HuggingFace Hub, Weights & Biases artifacts, or any other off-pod store during development; this is documented as a reproducibility limitation in Section 6.

## 6 CONCLUSIONS AND FUTURE WORKS

This report presented PathoLens-VLM, a research prototype for grounded whole-slide breast-cancer pathology report generation. During Design II the scope was narrowed from the five-module retrieval-based clinical decision support system in the interim report (Sections 1.4 and 4) down to one supervised mechanism the project could test end-to-end against an objective metric. The system connects a frozen CONCHv1.5 patch encoder, a trainable linear adapter and a Llama-3.2-3B-Instruct language model with LoRA through a single causal-attention sequence. About fifteen million trainable parameters sit on top of 3.3 billion frozen ones, and the full training run fits on a single 48 GB GPU in under twelve hours. The loss has three weighted terms: token cross-entropy on the response, KL divergence between per-sentence attention and a hybrid BCSS / CONCH-v1 grounding target, and an entropy-based faithfulness regulariser. A two-epoch warmup keeps early-stage grounding noise from disrupting caption learning.

On seven held-out test slides the prototype reaches BLEU-4 = 0.0559 and ROUGE-L = 0.1990 (sanity-check figures confirming coherent pathology-style generation). Its BCSS pointing-game accuracy is PG@5 = 0.081 against a uniform-random baseline of 0.019, a 4.4× lift over chance. Attention is far more concentrated than uniform (top-10 % mass 0.201, entropy ratio 0.938). Together these give feasibility-level evidence that a whole-slide pathology VLM can be trained to produce a measurable, non-random grounding signal: supervised text-to-vision attention can be aligned to BCSS-derived patch-level targets, and that alignment shows up on an objective metric on held-out data. The finding is promising but fragile. Pointing-game hits concentrate on a single slide, the test set is small, and the outputs are pathology-style structured reports, not clinically validated findings.

### 6.1 Limitations

PathoLens-VLM is an end-to-end research prototype. Its outputs are pathology-style structured reports produced by a language model fine-tuned on GPT-4-generated SlideInstruction captions; they are not clinically validated pathologist reports. No clinical expert has reviewed or verified any generated output. The following limitations apply to all claims in this report.

Small test set. Seven held-out slides and 37 concept-bearing sentences are enough to detect a non-random grounding signal, but too few to claim anything strong about generalisation or clinical correctness.

Fragile pointing-game signal. PG@5 hits concentrate on a single slide (TCGA-AO-A128). The aggregate 4.4× lift over uniform overstates the typical per-slide behavior; the other six test slides contribute zero hits at  $K \leq 10$ .

Supervision target noise. BCSS annotations cover only the annotated ROI tiles, not the entire slide. The CONCH-v1 pseudo-target depends on the zero-shot cross-modal alignment of the pretrained encoder; a preliminary study reported a zero-shot AUC-ROC of 0.608, so the pseudo-target is far from clean.

Attention is not causal explanation. The per-sentence attention from layer 14 is an attention-based grounding estimate; it does not show that the model actually relies on the highlighted patches to produce the sentence. Intervention-based faithfulness testing (zeroing top-K patches and measuring BLEU drop) is scaffolded in the repository but not yet executed at full scale.

Caption supervision quality. SlideInstruction captions are GPT-4-generated structured descriptions of TCGA reports. They have not been manually validated by a pathologist, and the model may have picked up GPT-4-style phrasing instead of clinically accurate diagnostic language.

Unvalidated attention layer. Attention layer 14 follows the LLaVA-style literature, but nobody has validated it on histopathology data; a layer sweep is listed as future work.

Lost runtime artifacts. The trained checkpoint from the fullscale run and the precomputed CONCHv1.5 / CONCH-v1 / BCSS embedding cache were stored only on the rental RunPod RTX A6000 pod. When the budget ran out at the end of the development window, the pod was decommissioned and its storage deleted; no copies were mirrored to HuggingFace Hub, Weights & Biases artifacts, or any other external store. The Streamlit app source code, all training and evaluation scripts, the YAML configs and the test suite are still in the GitHub repository, so the pipeline can be reproduced from a fresh GPU rental. The Streamlit demo, though, cannot be launched as-is: it needs embedding precompute (about five minutes per slide on an A6000) and at least a smoke training run (about one hour) to regenerate a checkpoint. This is an operational limitation, not a methodological one; a follow-up run would fix it by mirroring the checkpoint to HuggingFace Hub after each training cycle.

## 6.2 Future Work

Three immediate extensions are planned. The faithfulness intervention test scaffolded in `src/patholens/evaluation/intervention_test.py` will be run at full scale: zero out the top-K most-attended patches and measure the BLEU drop. A large drop would be direct behavioral evidence that the model relies on the highlighted regions. The keyword-based concept-recall evaluator (`concept_f1.py`) will be run on the full test set for per-class precision, recall and F1 against BCSS class coverage. A visual layer sweep across attention layers 8, 14 and 20 on three test slides will check the choice of layer 14.

Several longer-term directions follow. Scaling the training pool from 50 to 151 slides (the full BCSS-annotated TCGA-BRCA cohort) should cut the slide-to-slide variance of the pointing-game metric. Cross-dataset evaluation on CAMELYON16, deferred here because of its 150 GB footprint, would test generalization. Swapping the linear adapter for a Q-Former-style learned aggregator would test whether the simple adapter caps the grounding signal. Extending the system beyond breast cancer would need pixel-level annotations like BCSS for other tissue types; the BCSS-style approach should carry over to any pathology dataset with region-of-interest segmentation.

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